

535/1 PHYSICS

22th April

Paper 1

2025

2½ Hours

MANDELA COLLEGE SCHOOL

END OF TERM 1, S.4 2025

Paper 1

Theory

Duration: 2 Hours 30 Minutes

INSTRUCTIONS TO CANDIDATES:

This paper has two sections; A and B It has five examination items.

Each item carries 25 marks

Answer four items in all.

Any additional item(s) answered will not be scored.

All answers must be written in the answer sheets provided.

Please write well ,

A good handwriting will attract additional marks.

A poor hand writing will lead to loss of marks.

Item 1

A steel and tube industries company limited was contacted by the school to make a number of new items to restock their kitchen. Among the items on the invoice are; **saucepan, strainer, saucepan covers (lids), ladles, cups and plates for students**. The school did not specify the type of materials these items should be made of and therefore, the procurement officer of the company their raw materials supplier and the following was supplied following the invoice.

- ✓ Plain sheets of copper
- ✓ Plain sheets of aluminum
- ✓ Melamine resin powder
- ✓ Clay
- ✓ Wood

For the production team to come up with the items requested by the school, they have to subject these raw materials to either heat or force however these materials behave differently under these two factors.

Support knowledge.

-Heat capacity of copper is $400 \text{ J/kg} \cdot \text{K}$

- Heat capacity of aluminum is $900 \text{ J/kg} \cdot \text{K}$

Task

a) As a physics learner, with reason(s) help the production team to select the raw material suitable for each item to be produced. (15 marks)

b) Explain at particle level how these materials behave when subjected to either heat or when a force is applied. (10 marks)

Item 2



Your neighbor bought an electric kettle and connected it the first time to a power socket to boil 2 kg of water. Immediately, the neighbor heard a small explosion in the kettle and it stopped working.

The neighbor called you for help and on inspecting the kettle, it had the following features; a two pin plug connected to the kettle's cable. The cable had three wires in it, one with red insulation connected to the first pin, the second with blue insulation, connected to the second pin and the third had green insulation but not connected to any pin, also, the heating element had blown. The label on the kettle read 2 kW, 110 V, 50Hz.

Assist your neighbor to:

(a) know the name given to each of the wires in the cable.

(b) (i) understand explain the meaning of quantities written on the label.

- (ii) determine the resistance of the element of the kettle.
- (c) (i) understand the cause of blowing of the kettle.
- (ii) understand what could have been done to prevent blowing of the kettle.
- (iii) know other precautions that should always be taken in proper use of the kettle.

Task 3



In a certain home, the children's responsibility to draw water for cooking and also to boil it for drinking. They draw the water from an underground well using a rope and a bucket. The children have raised a complaint to their father that pulling water using a rope and a bucket is tiresome and are suggesting that a simple machine for drawing water be designed for them. Every day, they use an aluminium saucepan of mass 2 kg to boil 10 litres of water from a temperature of 24 °C to 90 °C. The father is planning to address his children's concern but is not certain of how to solve it, and is looking for guidance.

Task:

Using the knowledge of physics, help the father to

- (a) (i) design the simple machine that can be used and explain how it works.
- (ii) understand how to improve on efficiency of the machine designed in (a)(i).
- (b) understand how much heat energy is used daily by the family to boil drinking water.

Use;

Specific heat capacity of aluminum = $900 \text{ J kg}^{-1} \text{ } ^\circ\text{C}^{-1}$

Specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ } ^\circ\text{C}^{-1}$

Density of water - 1000 kg per cubic metres.

Item 4

A S.4 child and her mum have gone to buy some cooking utensils. The mum explains to the shop attendant that she usually gets problems in cooking beans since they take long to get ready when being boiled in water, leading to wastage of energy and time. The attendant shows her an appliance claiming it cooks fast due to its features. The lady is not convinced about the appliance and wishes to get more information before buying it; so she asks the child to explain more.

Task:

Assuming that the lady is your parent or guardian, use knowledge of Physics to educate her

Why;

- (i) the appliance makes the beans get ready faster. Page 10 VFAHS Phy Seminar '24
- (ii) the steel pan should have a plastic handle.
- (ii) the steel pan should have a thick body.
- (b) (i) why steel is an important property in this appliance.
- (ii) why the bottom of the pan and its sides are made of different colours respectively.
- (c) the different means by which heat from the heat source can reach the beans.

Item 5

You have been requested by your senior two brother to define and explain the following terms.

- ✓ Electric current.
- ✓ Electric charge.
- ✓ Resistance
- ✓ An electric circuit
- ✓ Potential difference.

Task

- a) Define , explain, and give the S.I. Unit of each of these quantities.
- b) Give the relationship between potential difference and electric current, and resistance and electric current.

I wish u Success

The end